Arduino driven 6x24 LED display

Wyatt Richard, Brendan Burdett

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Submitted to: Scott Chen

1. Project Overview

The project overall was a success. Every goal was met apart from a last moment issue with the PWM that we were unable to overcome in time. Many issues from software to hardware arose but all were overcome in time to meet our goals. The final prototype had access to the full alphabet, numeration and some punctuation as well as a rolling feature to display long messages, both Fahrenheit and Celsius temperature recordings and a simple string to control the message.

We are satisfied with the final prototype apart from the PWM issues that were not overcome.

2. Key Features

2.1. Scrolling 6x24 LED Display

The Display is composed of a six by twenty-four matrix of LEDs that displays text with a scrolling effect to allow for longer strings of text that would normally not fit on the smaller screen size. The text displayed on the screen is a string declaring the current temperature in the room in degrees Celsius with the brightness of the display adapting to the luminosity in the room dimming in darker rooms and increasing the brightness in lighter rooms. The output on the display is controlled by a combination of IC's and an Arduino UNO, the IC's are used to reduce the number of pins required from the Arduino from thirty down to seven pins. The IC's used are three 8-bit shift registers for the column control, and a demultiplexer and hex inverter for the row control. In order to display complex images on the screen the image must be displayed row by row, turning on one row at a time while the appropriate columns are also powered on. The rows are cycled through at a high enough speed that from a human's point of view all rows appear to be on at the same time. The adaptive brightness of the display is achieved through pulse width modulation, this is done by changing the pulse width of the signal sent to turn on the LEDs, with a longer pulse increasing the brightness and a shorter pulse decreasing it.

2.2. Temperature Sensors

The temperature measurement consisted of three temperature sensors. The single Celsius sensor was simply measured by the Arduino and stored as an integer for display. The two Fahrenheit sensors were connected to a custom-built operational amplifier circuit designed to have a gain of 10 for higher resolution on the data. Similarly, to how the Celsius reading is recorded the output of this operational amplifier was sent to the Arduino where the data was recorded for use if selected.

2.3. Luminosity Sensors

The luminosity measurement consists of two phototransistors measuring the ambient luminosity in the room which is then averaged to create a more accurate luminosity reading, this reading would then be used to determine the brightness of the LEDs on the screen. By constantly reading the luminosity the PWM would then be set to an appropriate brightness

based on tests. Due to the failed integration of the PWM, unfortunately the brightness value is unused. However, it is still recorded and stored if a user wished to display the value or for further attempt to integrate the PWM.

3. Key Engineering Challenges

One of the first challenges we ran into was with the row control of the display. In the Arduino we had the code cycling through rows one through six latching after all the rows had been cycled through, this caused the output to only be the last row all the way down, this was easily fixed by relocating the location of the latch to after each row. This challenge was also combined with the second challenge we faced which was that two of the rows were assigned to the incorrect pinout on the Arduino, leading to them being swapped on the display, once again this was easily fixed by switching the outputs to the correct pins resulting in the desired output on the display.

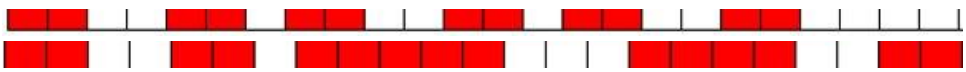


Image 1 & 2: Display output before and after fixing latch location

The next challenge we faced was the text of the display temporarily shifting upwards at the beginning of each loop of the shifting display, this problem was solved by reworking the conditions checked when shifting the display and removing the timer variable we were originally using for the shifting display.

Another challenge we faced was running out of global variable storage on the Arduino, when we ran into this problem, we had 39 separate arrays containing the data for each character that could be added to the output array this lead the Arduino being unable to run the code given to it. To overcome this we compressed the arrays by combining them into one array that contained reference numbers which called to a separate array holding all the possible rows that each character used, this method significantly decreased the amount of storage used by the Arduino allowing us to continue on with the project.



Image 1: Original array format



Image 2 & 3: Revised array format with reference key

The last challenge, which was unfortunately not overcome in time, was the pulse width modulation. Tested independently from the display with and without the temperature sensor the brightness of the LED was able to be controlled perfectly. When implementation came an unexpected issue arose. The PWM algorithm worked with the light sensor values as expected but when the value was used it did not work. The exact cause of the issue is still unknown however it is theorized that due to the refresh rate of the display the PWM was unable to operate correctly.

Due to issue with the LEDs being dulled (mistily due to being slightly burnt out), timing and no progress being made in the debugging the PWM feature was removed for the final prototype.

4. Future Proposed Features

One of the features we could implement is the use of an alternative display method, like a 7-segment display, this would allow for a smaller overall size as well as allow for more information to be displayed on the screen at one time.

Another feature we could implement is designing the PCB as an Arduino shield, which in addition to once again allowing for a smaller overall size, would also remove the need for all the jumper cables currently used connecting all the separate modules.

The next feature we could implement is designing and printing an enclosure for the project, having the project in an enclosure would make it significantly more presentable as well as protecting all the internal parts.

The last feature that we could implement is to add the ability for the user to manually control the brightness of the display as well as the ability for the user to swap between Celsius and Fahrenheit temperature readings.

Of the possible features listed, the only one we consider something that must be added is creating an enclosure for the device, if this device is to be commercially available then an enclosure is necessary as it not only protects the device from the user but also the user from the device. After the enclosure is implemented some of the features we should add are the Arduino shield and user input, although they are not necessary, they would add value to the product, the shield would lead to easier production by reducing the number of boards needed to one and removing all jumper wires. Adding user input to the device would allow for customization to the device depending on the user's needs, swapping between temperature measurement systems as well as controlling the brightness manually. Lastly the feature we could add is an alternate display, having a different display would allow for more information to be displayed at one time and remove the need for the information to be scrolled through, although this currently is not a problem it could definitely be improved upon

5. Prototype Development Summary

With the display already constructed, the very first phase in the project was interface with the display. This was done by studying the schematics offered and with a table of inputs that corresponded with a table of output that was previously created, an algorithm was created. Due to an issue in timing with the different latches there was an issue where the display would simply repeat the final row for every row. This was easily solved by moving the latch trigger

but led to another issue where the first and last row were in the wrong position. A simple adjustment to the algorithm overcame this issue.

After the display was operating Wyatt took over the main software and Brendan focused on the other features. The main software was further developed to accept a string which would be used by an algorithm to assemble the array needed for the output. A rolling feature was also later worked on and completed with a small issue that was overcome quickly.

The other features were tested individually using a separate Arduino Uno and separate components. Small, simplistic code was used to record the data from each sensor type for use. Once the data was successfully recorded a simple PWM program and circuit was constructed. Consisting of one LED the software cycled between 255 (max brightness) and 0 (off) back and forth. This program was later introduced into a separate program which incorporated the readings from the luminosity sensor and would adjust the brightness accordingly. With the additional features built and the main code constructed the operational amplifier was next.

The first phase was designing a virtual circuit in multisim which was completed rapidly with the resources given. A gain of 10 was decided upon due to the resistors available and that the number was easily multipliable and divisible to allow for easier testing. The multisim schematic was then used to construct the breadboard prototype. Other than a small mistake in the grounding of the operation amplifier and the power supplies there were no issues. The operational amplifier operated exactly as expected and was complete, leaving the integration of everything as the final step.

When implementation was about to begin there was an issue with the overall size of the program and the variables. The Arduino ran out of storage. This issue was overcome using an idea developed in the earliest stages of the main program where the letters would share similar parts which could be assembled to construct a full letter. With the algorithm adjusted and the variables adjusted to overcome this issue implementation began.

The implementation of the sensors had no issues. They were simply connected to the appropriate pins and with the code developed in the individual test implemented correctly they worked with no issues. At that point in time the project was nearly completed but then there was an issue with the PWM implementation. The algorithm operated exactly as expected but the display did not operate as expected. It was anticipated that, like in the single test, the HIGH of the correct output would be replaced with the calculated value for the PWM. This did not work. Despite the efforts the implementation of the PWM failed. Although the cause is uncertain, it is theorized that it may have had something to do with the displays own timing and the PWMs timing not matching. Given more time this issue would most likely have been solved but due to the issue being discovered so late it was not fixed and PWM was removed from the final prototype.

Throughout the project ideas were shared and the group collaborated well to create the final prototype. Overall, the final prototype was fully operational and met all goals with the only exception being the PWM issues.